
The Shape of a Knowledge Boundary: A Transformer Trained Only on Pre-1939 Text

Junlei An
1010864165

<https://github.com/Shinoaki798/1939> I used Claude Fable 5.1 to do the proposal.

Introduction

A language model’s knowledge cutoff is usually described as a wall. I suspect it is a **membrane with structure**: some post-cutoff concepts are unreachable, others are assembled from parts the model already has and are half predictable. To characterise it I will train **one Transformer network** from random initialisation on newspaper text ending before the Second World War — American and German press, the German in the original, because the country that started the war kept a record the American press could not — and use that one model to answer three questions. **RQ1 (permeability)**: which classes of post-1939 concepts can a pre-1939 model reach? **RQ2 (invertibility)**: can the boundary be inverted into a tool that dates a passage and audits a corpus for post-1939 text? **RQ3 (foresight by perspective)**: how much of 1939–45 was already implied by the press of mid-1939 — American or German? Beyond history, this bounds what “held-out” tests test and whether contamination is detectable. Deep learning is the instrument: these quantities exist only in a trained model’s next-token distribution, so the project *is* the training, validation and testing of that model. Training ends **30 June 1939**; July–August 1939 is an **embargo**, excluded from training and evaluation.

Illustration

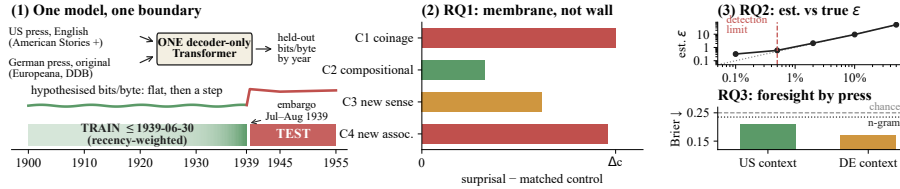


Figure 1: One model, three measurements. Shapes are hypothesised, not results.

Background & Related Work

Language models degrade on text written after their training window [12]; temporal conditioning improves future-fact calibration [3]; Time Machine GPT [4] trains point-in-time models. Recent from-scratch models with 19th- and 20th-century cutoffs, including 1939 [7, 8, 17], aim to *produce* such models; this project studies the boundary itself. SemEval-2020 Task 1 [18], HistBERT [16] and 19th-century models [10] measure how *existing* words drift with masked encoders; RQ1 asks which *not-yet-existing* concepts a fixed earlier vocabulary can reach, which needs a causal model with a hard boundary. ChroniclingAmericaQA [15] shows OCR errors cut accuracy, hence an OCR covariate.

Data Processing

Sources. All text is pre-cutoff, in the original language; nothing is machine-translated. *English*: American Stories [1], article-level text from Chronicling America, is the backbone and sole source of scored sets; training-only additions: Chronicling America pages digitised since that snapshot (1930s,

page OCR) [14], the Congressional Record [6], the Federal Register [19] and case law [9] (legal text $\leq 10\%$), and public-domain books [13] ($\leq 12\%$). *German*: newspapers from Europeana [5] and the Deutsches Zeitungsportal [2] (38 titles in 1933–39), deduplicated by title and date, plus the 1925 and 1930 issues of the NSDAP daily *Völkischer Beobachter* [11]; German is $\leq 25\%$ of seen tokens per period. Every file is pinned by revision and checksum. **Recency weighting**. The pool centres on the 1910s but the model must live in 1939: the 1930s are seen twice; the 1920s fill the remainder ($\leq 35\%$ of seen tokens); pre-1920 $\leq 8\%$; within a period, sampling weight $\propto \exp(-(1939 - \text{year})/5)$. **Splits** by publication date, never shuffled: a fixed 2% of every year 1900–1955, per language, is held out and never trained on; TRAIN = the rest $\leq 1939-06-30$; VAL = a disjoint 2%; TEST-A/B/C = held-out 1930–38 / 1939-09–45 / 1946–55. **Cleaning**, in order: (1) language filter (stopword vote per article and title-year); (2) MinHash dedup of reprints (word 5-grams, Jaccard ≥ 0.8) before splitting; (3) OCR quality = period-lexicon hit rate per language, cut at a threshold set from its histogram; (4) documents containing a C1 coinage (English or German list) are dropped whole. **Format**: 48k BPE trained only on native pre-cutoff English and German; documents joined by an end-of-text token, packed into 2048-token sequences; all numbers are bits-per-byte. **RQ1 probe set**: ~ 200 post-cutoff terms in four classes — C1 coinage (*radar*), C2 compositional (*atomic bomb*), C3 old word, new sense (*blitz*), C4 proper noun, new association (*Hiroshima*) — each with a frequency-matched control in ≥ 5 sentence contexts, frozen in advance. **RQ2 calibration**: held-out pre-cutoff corpora with a known fraction ε (0.1–50%) of 1939-09–1945 articles mixed in. **RQ3 propositions**: ~ 60 binary propositions about 1939–45, each with same-topic US (English) and German (German) contexts from June and late August 1939 (embargo text as conditioning only); continuations are English.

Architecture

One **decoder-only Transformer** as in the course: token embeddings, stacked blocks of multi-head self-attention and position-wise feed-forward layers, residual connections, layer normalisation, softmax over the vocabulary, with attention **causally masked** (position i sees only $\leq i$), a next-token predictor. Decoder-only because every measurement is a sequence probability of unseen text: a masked encoder’s pseudo-likelihoods give none, and an encoder–decoder has no source sequence here. From scratch in PyTorch, no pretrained weights, **no second Transformer**: all controls are built at evaluation time on this one model. **Size**: 350M parameters (24 layers, $d=1024$, 16 heads, 48k vocab) on up to 10B seen tokens (≈ 29 /parameter); no text is seen more than twice, as repetition raises memorisation and biases surprisal. Compute $\approx 2.1 \times 10^{19}$ FLOPs, ~ 85 h on the RTX 5080 (H100: 15 h), before the midterm. **Measurement**. For term class c , $\Delta_c = \mathbb{E}_{t \in \mathcal{T}_c}[s(t)] - \mathbb{E}_{t \in \mathcal{C}_c}[s(t)]$, s the surprisal in fixed contexts, $\mathcal{C}_c$ matched controls. RQ1 is the profile ($\Delta_{C1..C4}$); RQ2 fits a monotone map to ε and reports the minimum detectable ε ; RQ3 scores forced-choice continuations by log-likelihood ratio as Brier score per perspective; all with bootstrap 95% CIs. **Success**: Δ_{C2} significantly smallest; a step at 1939, not a slope; $\hat{\varepsilon}$ monotone; perspectives above the n -gram floor and significantly different.

Baseline Model

Following the course progression, with the same split, tokenizer and budget: a **vanilla RNN**, an **LSTM** and a **GRU** language model (2 layers, ~ 50 M parameters, next-token objective), which justify the Transformer as primary model. A **modified Kneser–Ney 5-gram** (KenLM defaults) is the non-neural floor; it cannot compose unseen compounds, so it should show no C1/C2 ordering. **Plan** (individual): corpus frozen 20 Oct; baselines 17–21 Oct; Transformer 22 Oct–1 Nov; then evaluation.

Ethical Considerations

Pre-1939 newspapers carry the racial, ethnic and gender language of their time; the model reproduces it, which is documented, not sanitised. German press after 1933 was state-controlled, so the “German perspective” is what Germans were permitted to read; RQ3 results describe a record, not nations, and corpus shares by source are reported. In collection, OCR noise concentrates in older and German (Fraktur) text, and year-only book dates can mislead (hence the C1 screen). All sources are public domain or openly licensed, and attributed; as the papers name private people, the model is not released for querying them; the RQ2 tool, misusable as an accusation, ships with its detection limit.

References

- [1] M. Dell, J. Carlson, T. Bryan, E. Silcock, A. Arora, Z. Shen, L. D’Amico-Wong, Q. Le, P. Querubin, and L. Heldring. American stories: A large-scale structured text dataset of historical U.S. newspapers. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.
- [2] Deutsche Digitale Bibliothek. Deutsches Zeitungsportal. <https://www.deutsche-digitale-bibliothek.de/newspaper>, 2024. Hugging Face mirror: storytracer/German-PD-Newspapers.
- [3] B. Dhingra, J. R. Cole, J. M. Eisenschlos, D. Gillick, J. Eisenstein, and W. W. Cohen. Time-aware language models as temporal knowledge bases. *Transactions of the Association for Computational Linguistics*, 10, 2022.
- [4] F. Drinkall, E. Rahimikia, J. B. Pierrehumbert, and S. Zohren. Time machine GPT. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 2024.
- [5] Europeana Foundation. Europeana Newspapers: Full-Text Collection. <https://www.europeana.eu/en/collections/topic/18-newspapers>, 2015. Hugging Face mirror: biglam/europeana_newspapers.
- [6] M. Gentzkow, J. M. Shapiro, and M. Taddy. Congressional Record for the 43rd–114th Congresses: Parsed Speeches and Phrase Counts. Stanford Libraries, 2018.
- [7] D. Göttlich, D. Loibner, G. Jiang, and H.-J. Voth. Ranke-4B: History LLMs with fixed knowledge cutoffs. <https://github.com/DGoettlich/history-llms>, 2025.
- [8] H. Grigorian. TimeCapsuleLLM: A language model trained only on data from 1800–1875. <https://github.com/haykgrigo3/TimeCapsuleLLM>, 2025.
- [9] Harvard Law School Library. Caselaw Access Project. <https://static.case.law/>, 2024. Hugging Face mirror: free-law/Caselaw_Access_Project.
- [10] K. Hosseini, K. Beelen, G. Colavizza, and M. Coll Ardanuy. Neural language models for nineteenth-century English. *Journal of Open Humanities Data*, 7, 2021.
- [11] Internet Archive. Völkischer Beobachter, 1925 and 1930 issues. <https://archive.org/details/pub-volkischer-beobachter>, 2024. Digitised from microfilm.
- [12] A. Lazaridou, A. Kuncoro, E. Gribovskaya, D. Agrawal, A. Liska, T. Terzi, M. Gimenez, C. de Masson d’Autume, T. Kocisky, S. Ruder, D. Yogatama, K. Cao, S. Young, and P. Blunsom. Mind the gap: Assessing temporal generalization in neural language models. In *Advances in Neural Information Processing Systems*, 2021.
- [13] Library of Congress. Library of Congress Public Domain Books. <https://www.loc.gov/collections/selected-digitized-books/>, 2024. Hugging Face mirror: storytracer/LoC-PD-Books.
- [14] Library of Congress. Chronicling America: Historic American Newspapers. <https://chroniclingamerica.loc.gov/data/ocr/>, 2026. Batch OCR, pages added after the American Stories snapshot.
- [15] B. Piryani, J. Mozafari, and A. Jatowt. ChroniclingAmericaQA: A large-scale question answering dataset based on historical American newspaper pages. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.
- [16] W. Qiu and Y. Xu. HistBERT: A pre-trained language model for diachronic lexical semantic analysis. *arXiv preprint arXiv:2202.03612*, 2022.
- [17] A. Radford, N. Levine, and D. Duvenaud. Introducing talkie: A 13B vintage language model from 1930. <https://talkie-lm.com/introducing-talkie>, 2026.

- [18] D. Schlechtweg, B. McGillivray, S. Hengchen, H. Dubossarsky, and N. Tahmasebi. SemEval-2020 task 1: Unsupervised lexical semantic change detection. In *Proceedings of the 14th International Workshop on Semantic Evaluation*, 2020.
- [19] U.S. Government Publishing Office. Federal Register, 1936–1939. <https://www.govinfo.gov/app/collection/fr>, 2024.